



Tribhuvan University
Faculty of Humanities and Social Sciences
Ratna Rajyalaxmi Campus
Mid-Term Examination 2024

Bachelor in Computer Applications
Course Title: Probability and Statistics
Code No: CAST202
Semester: III

Full Marks: 60-40
Pass Marks: 24-16
Time: 3 2 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

(4×5 = 20)

Attempt any FOUR questions:

2. Define statistics. Write the importances and limitations of statistics.
3. What do you mean by descriptive and inferential statistics? Write with examples. Write the scopes and applications of statistics in computer applications.
4. Calculate Q_1 , Q_3 , D_6 and P_{80} of the following data:

Items	0-10	10-20	20-30	30-40	40-50
Frequency	4	7	10	6	5

5. Calculate the mean, median and mode of the following data:

Age(in year)	5-14	15-24	35-44	45-54	55-64
No. of persons	3	8	11	6	2

6. From the following data, calculate the Pearson's correlation coefficient and interpret the result:

X	20	30	40	10	50
Y	15	12	9	18	13

7. Following data shows the age in month and weight in kg of newly born baby:

Age	0	3	5	6	8
weight	2.5	4	5.5	8	10

Estimate the weight of the baby at the age of 4 months.

8. What do you mean by the binomial distribution? The mean and variance of a binomial distribution are 8 and 4 respectively. Find the probability of exactly 2 successes.

Group C

(1×10 = 10)

Attempt any ONE question:

9. From the following information given in the table:

	Factory A	Factory B
Average weekly wages(Rs.)	560	590
Standard deviation	60	50
No. of workers	100	80

- (i) Which factory pays larger amount as weekly wages?
- (ii) Which factory has greater variability in the distribution of the weekly wages?
- (iii) What is the mean and standard deviation of all the workers in two factories taken together?
- (iv) Find the coefficient of variation of 180 workers taken together of the factory?
10. Two brands of computer are tested for their life and following results were obtained:

Life(in year)	4-6	6-8	8-10	10-12	12-14
Brand A	12	15	18	14	8
Brand B	8	12	15	10	5

Both brands are offering the same price and advertising in favor of their brands saying that the brand has consistent life. If you are required to purchase computer of one of these two brands, which computer would you prefer and why?

11. State the addition law and multiplication law of probability. What do you mean by the conditional probability? Given that if 25% students passed in English, 20% passed in Nepali and 10% passed in both subjects. A student is selected at random. Find the probability that
 - (i) he has passed in Nepali, it is being given that he has passed in English
 - (ii) he has passed in English, it is being given that he has passed in Nepali
 - (iii) he has passed in English or Nepali.

Best of Luck



Tribhuvan University
Faculty of Humanities and Social Sciences
Ratna Rajyalaxmi Campus
Mid-Term Examination 2025

Bachelor in Computer Application

Course Title: Data Structure and Algorithm

Code No: CACS201

Semester: III

Full Marks: 60-40

Pass Marks: 24-16

Time: 3 2 hours

Candidates are required to answer the questions in their own words as far as possible.

Group – A
Group B

Attempt any six [5 x 6 = 30]

2. What is Data Structure? Convert $a\$b*c-d+e/f/(g+h)$ into postfix expression.
3. What is singly linked list? Write an algorithm to add a node at the beginning and end of singly linked list.
4. What is binary search? Write an algorithm to search an item using binary search with appropriate illustration.
5. Define circular queue. Write an algorithm to insert an item in a circular queue.
6. What is recursion? Write an algorithm to solve Tower of Hanoi problem.
7. Describe Dijkstra's shortest path algorithm with suitable illustration.
8. What is hashing? Describe the types of collision resolution techniques.
9. Explain the different traversal techniques used in binary tree with examples.

Group C

Attempt any two [2 x10 = 20]

10. What is stack? List the application of the stack. Write an algorithm to perform PUSH and POP operation in stack.
11. What is heap? Explain quick sort algorithm with Big-oh notation in best case, average case and worst case and trace it to sort the data: 8, 10, 5, 12, 14, 5, 7, 13.
12. Explain graph data structure with its applications. Describe Floyd Warshall Algorithm with example.



Tribhuvan University
Faculty of Humanities and Social Sciences
Ratna Rajyalaxmi Campus
Mid-Term Examination 2025

B

Bachelor in Computer Applications

Course Title: OOP in Java

Code No: CACS 204

Semester: III

Full Marks: 60-40

Pass Marks: 24-16

Time: 3 2 hours

Candidates are required to give their answer in their own word as far as practicable.
The figures in the margin indicates full marks.

Group B

Attempt any FOUR questions.

[4×5= 20]

11. Write a program to take a string from user and display whether the length of the string is even or odd. [5]
12. How for loop and for-each loop in Java is different? Write a program having an array of 6 floating point numbers and display all the elements of the array using for-each loop. [2+3]
13. Why constructors are used in a class? Write a program to demonstrate parameterized constructor. [2+3]
14. What is method overloading? Write a program having a class *AreaCalculator* with overloaded methods named *area()* to compute the area of: a circle (given radius), a rectangle (given length and breadth) and a square (given side length) [1+4]
15. Write a Java program to create a base class *Vehicle* with attributes brand, speed, and type, and a method called *describe()* that prints out its brand and type. Inherit the class in *Car* and *Bike* classes and override the *describe()* method to print specific messages for each vehicle. [5]

Group C

Attempt any ONE question

[1×10 = 10]

16. How static polymorphism and runtime polymorphism is different? Write a program considering following scenario:
You are tasked with developing a simple Ride Booking System like 'Pathao' that allows users to choose between different types of rides, such as a Bike, Car, or Scooter. Each ride type charges a different fare per kilometer: Rs. 16/km for Bike, Rs.55/km for Car, and Rs. 14/km for Scooter. To implement this, define an interface named *IRide* that contains a method *calculateFare(double distance)*. Then, create three classes—*Bike*, *Car*, and *Scooter* each of them should implement the *IRide* interface and provide their own implementation of the *calculateFare* method. In your *main()* method of main class, demonstrate runtime polymorphism by using a *IRide* reference to point to different ride types based on the user's input, and calculate the fare for a given distance. This program should simulate how a real-world ride-booking app calculates fares dynamically based on the selected vehicle and distance traveled. [3+7]
17. Compare between concrete class, abstract class and interface in Java. Explain about exception and exception handling in Java with the list of associated keywords. Explain the use of try, multiple catch and finally blocks with suitable programming example. [3+3+4]

Sec-B



Tribhuvan University
Faculty of Humanities and Social Sciences
Ratna Rajyalaxmi Campus
Mid-Term Examination 2025

SEC-B

Bachelor in Computer Applications
Course Title: Web Technology
Code No: CACS 205
Semester: III

Full Marks: 60-40
Pass Marks: 24-16
Time: 3 2 hours

Candidates are required to answer the questions in their own words as far as possible.
Group B

Attempt any SIX questions. [6×5 = 30]

2. What is Hyperlink object? Describe different type of Hyperlink with example.
3. Explain what are the tags and attributes for Form in HTML document?
4. What is anonymous access? Discuss about Integrated windows authentication
5. Explain tire technology with examples?
6. Define session. Explain its use with example
7. What is CSS? Discuss CSS Selector with example.
8. Write short note on following:
 - \$ _POST
 - rowspan and colspan

Group C

Attempt any TWO questions. [2×10= 20]

9. Differentiate between tags and attributes? Write a HTML and CSS code design a data entry form for marks entry of a student using different html elements like (table, image, formatting tags, links).
11. Discuss Array in php. Develop a simple web page that asks the users input (Product Name, Rate, Quantity, Supplier, date) and display it to another page Server-side script.

SEC-B



Tribhuvan University
Faculty of Humanities and Social Sciences
Ratna Rajyalaxmi Campus
Mid-Term Examination 2025

Bachelor in Computer Applications

Course Title: System Analysis and Design

Code No: CACS203

Semester: III

Full Marks: 60-40

Pass Marks: 24-16

Time: 3 2 hours

Candidates are required to answer the questions in their own words as far as possible.

Group – A

Short Answer Questions 4*5=20

Attempt any 4 questions.

1. Compare and contrast the scheduling tools: PERT and Gantt Chart.
2. What are the potential sources of software needed by an organization?
3. Describe the various types of information systems.
4. Identify and explain the common skills and tasks of a project manager.
5. Describe value chain analysis and how organizations use this technique to evaluate and compare projects.
6. List and discuss the different types of project feasibility factors.

Group C

Long Answer Questions 1*10=20

Attempt any 1 question

7. Assuming monetary benefits of an information system at \$85,000 per year, one-time costs of \$75,000, recurring costs of \$35,000 per year, a discount rate of 12 percent, and a five-year time horizon, calculate the net present value of these costs and benefits of an information system. Also calculate the overall return on investment of the project and then present a break-even analysis. At what point does breakeven occur.
8. Describe four traditional techniques for collecting information during analysis. When might one be better than another?